

GROWTH AT RISK AND MACROECONOMIC UNCERTAINTY: EVIDENCE FROM MEXICO

CRECIMIENTO EN RIESGO E INCERTIDUMBRE MACROECONÓMICA: EVIDENCIA PARA MÉXICO

Alfredo Salgado

Banco de México

<https://orcid.org/0009-0000-5304-4208>
asalgadot@banxico.org.mx

Alejandro Trujillo

Banco de México

<https://orcid.org/0009-0006-7892-3341>
atrujillo@banxico.org.mx

Resumen:

Analizamos la relación entre la incertidumbre y las expectativas de crecimiento económico en México mediante la metodología de Crecimiento en Riesgo. En primer lugar, estimamos una regresión por cuantiles del crecimiento anual del producto, condicionada a los valores rezagados de una medida de incertidumbre macroeconómica y otros determinantes. Posteriormente, con base en los valores ajustados de dicha regresión por cuantiles, estimamos los parámetros de una distribución t-asimétrica del crecimiento económico esperado. Nuestros resultados muestran que un aumento en la incertidumbre incrementa de manera significativa la probabilidad de observar resultados de crecimiento más bajos. Estos hallazgos son robustos frente a diversas medidas de condiciones financieras, incertidumbre de política económica, exposición al riesgo y actividad económica.

Abstract:

We analyze the relationship between uncertainty and economic growth expectations in Mexico through the Growth at Risk methodology. First, we estimate a quantile regression of annual output growth conditional on lagged values of a measure of macroeconomic uncertainty and other drivers. Then, based on the fitted values of the quantile regression, we estimate the parameters of a t-skewed distribution of expected economic growth. Our results show that increased uncertainty significantly increases the probability of observing lower growth outcomes. These results hold under various measures of financial conditions, policy uncertainty, risk exposure, and economic activity.

Clasificación JEL/JEL Classification: C53, E23, E27, E32, O40.

Palabras clave/keywords: macroeconomic uncertainty, financial conditions, growth at risk.

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1. Introduction

In recent decades, the Mexican economy has faced episodes of high uncertainty from both internal and external factors. Given the magnitude of these uncertainty shocks and their potential link to growth at risk and vulnerability of growth, it is relevant to analyze their impact on growth expectations empirically. This paper contributes to the growing literature on the relationship between macroeconomic uncertainty and economic growth expectations. Our analysis is carried out by means of the Growth at Risk (GaR) methodology proposed by Adrian *et al.* (2019)¹ applied to the case of Mexico.

Related literature, primarily focused on the United States and emerging economies like China, has emphasized the role of uncertainty as an key driver of economic growth and growth expectations (Bloom, 2009; Bloom, 2014; Bonciani and Jason, 2019; Jovanovic and Ma, 2020; Gu *et al.*, 2021). According to those studies, high uncertainty can negatively affect macroeconomic, microeconomic, and financial decisions as economic agents adopt precautionary behaviors, delaying consumption and investment plans while waiting for better conditions. Examples of the effects include:

- Firms may postpone their investment and hiring plans due to the expectation of increasingly uncertain returns.
- Households may raise their precautionary savings, resulting in reduced consumption.
- Financing costs may increase driven by the expectation of higher risk premia.

¹ The main strength of GaR methodology is its ability to assess the entire distribution of expected GDP growth and monitor the evolution of risks to economic activity over time. By using GaR analysis, policymakers can quantify the likelihood of risk scenarios, which would serve as a basis for preemptive action (Prasad *et al.*, 2019). Another tool usually used to analyze expected growth distributions is the 'fan charts' that convey uncertainty around central projections from linearised macroeconomic models. Recent literature has shown that GaR models provide improved estimates of the left tail of the GDP growth distribution, particularly at medium-term horizons, compared to the GDP fan charts. However, GaR models generally perform worse than the fan charts at the center of the distribution (See Lloyd *et al.*, 2022). In this way, it has been shown that by combining both methodologies, a better projection of the complete distribution of expected GDP growth could be obtained.

- The responsiveness to interest rate changes may decline, thereby reducing the effectiveness of monetary policy.²

It seems plausible that increases in uncertainty not only affect point estimates of forecasted economic growth, but also its entire probability distribution,³ potentially biasing expected growth (Adrian *et al.*, 2019; Prasad *et al.*, 2019). Adrian *et al.* (2019) pioneered the use of the GaR methodology to model United States gross domestic product (GDP) growth expectations based on economic and financial conditions, using the National Financial Conditions Index (NFCI), published by the Federal Reserve Bank of Chicago. They found that the estimates of the first quantiles of the GDP growth distribution are more sensitive to the state of financial conditions than central moments or upper quantiles. In addition, they proposed some measures to quantify the downward and upward vulnerability of expected GDP growth.

For the case of Mexico, some studies like Banxico (2019) have examined the impact of financial conditions on growth distribution.⁴ Particularly, Banxico (2020a) focuses on the period of the COVID-19 pandemic and found that the estimated distribution of conditional growth measured by the Global Indicator of Economic Activity (IGAE) shifted left due to tighter financial conditions and a slowdown in economic activity.

Unlike the traditional GaR approach, which focuses on financial conditions, we examine how macroeconomic uncertainty affects growth expectations. Our approach is based on two observations: First, financial variables may not fully capture expected growth. In

² Aquino *et al.* (2022) analyze the impact of monetary policy in the United States under high and low uncertainty regimes. Findings indicate that in low uncertainty periods, the United States monetary policy is effective since it drops economic activity and inflation. In contrast, in the high uncertainty regime, the United States monetary policy loses its effectiveness since it does not affect, or has a reduced effect, on economic activity and inflation.

³ For instance, policymakers, such as central banks and multilateral international organizations, such as the International Monetary Fund and the World Bank, provide not only point estimates of the mean or median GDP growth and other economic variables of interest but also a detailed analysis of the balance of risks around a central scenario, thereby emphasizing the importance of the shape of conditional growth distribution.

⁴ These analyses use a non-public financial conditions index estimated by the Bank of Mexico.

a recent paper, Plagborg-Møller *et al.* (2020) show evidence that financial variables have limited predictive power for the distribution of United States GDP growth over short horizons, especially, but not limited to, the underlying risk in the left tail of the distribution. Second, tighter financial conditions may reflect rising uncertainty, as greater uncertainty can lead to more restrictive credit and liquidity conditions, affecting economic performance.⁵

Several papers have analyzed the impact of uncertainty on macroeconomic variables in Mexico. However, to the best of our knowledge, none have examined the impact of uncertainty on the complete distribution of expected economic growth. For instance, López and Bush (2019) investigate the effect of uncertainty on exchange rate volatility for 1999-2018 in Mexico using survey-based indicators of uncertainty. They found that greater uncertainty increases exchange rate volatility, particularly during recessions. Catalán (2019) analyzes the impact of uncertainty shocks on economic activity in Mexico. He showed that economic policy uncertainty (EPU) increases prices and interest rates while reducing investment and employment. Finally, Cebreros *et al.* (2020) analyze the impact of trade policy uncertainty (TPU) on foreign direct investment (FDI). The authors found that TPU negatively affected FDI, especially in export-oriented states. Given that uncertainty seems to be an important determinant of economic activity, we analyze its impact on the full distribution of growth expectations in Mexico.

To quantify uncertainty levels in Mexico, we use the Macroeconomic Uncertainty Index (MUI), estimated by the Bank of Mexico based on Jurado *et al.* (2015).⁶ The MUI reflects the average difficulty in predicting the economic conditions, with higher values indicating increased uncertainty. We apply a GaR analysis to relate the annual growth rate of the IGAE⁷ to the MUI and other growth determinants

⁵ In this regard, it can be shown that the NFCI and a financial conditions index for the Mexican economy proposed by Carrillo and García (2021) had a high positive correlation since 2009, which could be a reflection of the effect of external uncertainty on financial conditions in Mexico.

⁶ For more details about the methodology to construct the MUI for Mexico and the variables included in its estimation, see Banxico (2020b).

⁷ The IGAE is a monthly series that measures the real sector of the economy. Being highly correlated with GDP, it is sometimes referred to as the monthly GDP. This variable is used for our main analysis to better capture the state and variations of uncertainty in the economy, since the MUI is a monthly indicator. Section 3.5 presents the results of our estimates based on the GDP of Mexico.

such as the United States industrial production index (IPUS) and inflation.⁸ From now on, these estimates of IGAE will be referred to as expected IGAE. Following Adrian *et al.* (2019), we perform a two-stage estimation. First, we estimate a quantile regression to assess the impact of uncertainty along the growth distribution quantiles.⁹ Then we fit a *skewed-t* distribution to these estimates, allowing us to have a detailed analysis of growth dispersion, bias, and the probability of different growth outcomes under uncertainty.

Our main results suggest that higher uncertainty increases downside risks to Mexico's expected growth. The lower quantiles estimates of the expected IGAE growth distribution show greater variability with increases in uncertainty levels compared to the central moments and the right tail. Furthermore, during economic crises, this approach seems particularly suitable for analyzing recession risks due to uncertainty spikes. These results align with Jovanovic and Ma (2020), who observed similar effects for the United States, where higher macroeconomic uncertainty is associated with a more spread-out distribution of expected output growth and significantly impacts the lower tail. Similarly, Gu *et al.* (2021) found that economic policy uncertainty in China alters the skewness and kurtosis of the expected growth distribution, consistent with our model.

Our baseline results are robust when controlling for alternative measures of uncertainty, both internal and external, and for the addition of variables that measure risk and financial conditions in Mexico. In particular, estimates of the impact of uncertainty remain significant when controlling for the United States NFCI, the Mexican financial condition index proposed by Carrillo and García (2021), and differ-

Our main conclusions remain unchanged.

⁸ Although the MUI includes variables related to economic activity, we consider it unlikely to present significant endogeneity with the IGAE. First, the MUI is based on the average forecast errors of over 100 variables, making it unlikely that the IGAE would meaningfully impact these errors. Second, to address potential endogeneity, we include controls in our quantile regressions, such as IGAE lags, the exchange rate, inflation, interest rates, and the United States industrial production, which are key determinants of Mexico's economic activity and help mitigate endogeneity issues.

⁹ Unlike ordinary least squares estimates, quantile regression is estimated at the different quantiles of the distribution of the dependent variable and not just at a central moment, so it is more robust to the presence of extreme values. See Adrian *et al.* (2019) and Koenker and Bassett (1978) for more details about quantile regression methods.

ent EPU measures (Mexican, global, and the United States). We find similar results when we use traditional risk measures, such as changes in the real and nominal exchange rates, and risk premium measures. Robustness tests with six alternative economic activity indicators - gross fixed investment, private consumption, industrial activity, tertiary sector IGAE, formal employment (proxied by the number of jobs affiliated to the IMSS),¹⁰ and the employment rate- also show that uncertainty still impacts expected economic activity similarly to the case of IGAE.¹¹

Finally, we provide a detailed analysis of the impact of uncertainty on the distribution of economic growth during the COVID-19 pandemic. Our findings show that, in times of extreme uncertainty like economic crises, the estimated conditional distribution on MUI aligns more closely with actual output growth. High MUI levels exacerbate the negative bias in growth projections, making the central moments of the estimated distribution relatively optimistic compared to realized values.

The rest of the paper is organized as follows. In section 2, we make a first approach to the relationship between uncertainty and economic activity in Mexico through a vector autoregressive analysis (VAR) that shows the relative importance of the MUI shocks on the dynamics of economic growth in Mexico. In section 3, we present the methodology, general results, and robustness exercises of our GaR analysis between macroeconomic uncertainty and IGAE growth distribution. In section 4, we analyze the results of our estimates for the particular case of the COVID-19 pandemic. Finally, in section 5, we present some general conclusions.

2. Uncertainty and economic activity in Mexico

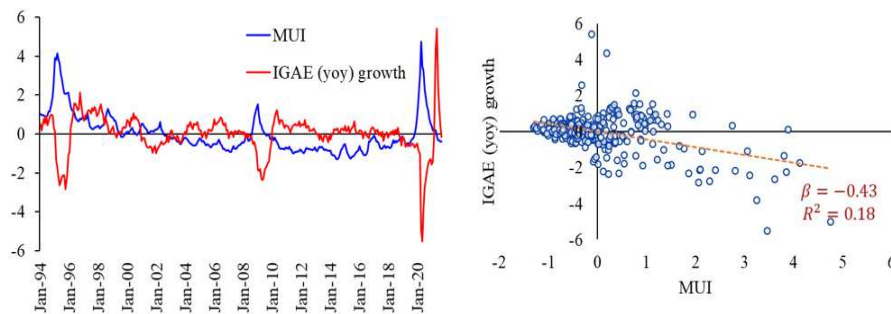
The purpose of this paper is to analyze the empirical relationship between economic growth and uncertainty since, from a theoretical

¹⁰ *Instituto Mexicano del Seguro Social*, the social security institute of Mexico.

¹¹ While our estimates of the MUI's impact on growth expectations are significant and robust, the MUI has a timeliness issue compared to other uncertainty, risk, and volatility measures. This lag is due to delayed publication of some input indicators and the loss of at least one observation in calculating forecast errors (Jurado *et al.*, 2015). Consequently, the timeliness of the MUI's impact on growth expectations is limited. Future research may focus on developing a more timely MUI, though initial tests show that faster estimates come with increased volatility, potentially overstating the MUI's effect on growth expectations.

and empirical point of view, it has been found that uncertainty is an important determinant of growth.¹² For our estimates, throughout the paper, we consider a sample from July 1993 to December 2021. As shown in Figure 1, during this period, sharp drops in economic growth are generally accompanied by high levels of macroeconomic uncertainty.

Figure 1
*IGAE (yoy) growth and macroeconomic
 uncertainty index for Mexico*
April 94 - December 21



Notes: To facilitate comparison, both series are standardized.

Source: Authors' elaboration.

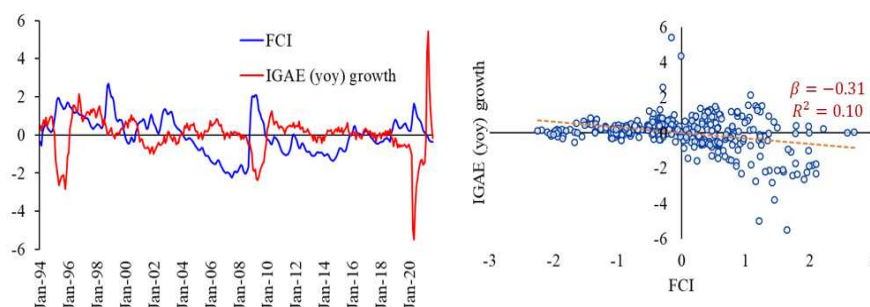
Several studies have analyzed the impact of uncertainty on growth through VAR models (see Baker *et al.*, 2016, 2020; Bloom, 2009; Bonciani and Jason, 2019; ECB, 2016; Gieseck and Rujin, 2020). This type of analysis combines real, nominal, and financial variables. These indicators are related to the level of economic activity, financial conditions, the level of uncertainty, and the exposure of the economy to risk. Financial conditions could also be relevant to analyze growth expectations (Adrian *et al.*, 2019; Banxico, 2019, 2020a).

To analyze the impact of uncertainty on growth, controlling for financial conditions in Mexico, we include in our analysis the Finan-

¹² One possible concern is that the correlation may arise from including the IGAE in the MUI. However, this is unlikely. The MUI is based on 125 equally weighted economic series, giving the IGAE a minimal weight. Additionally, the MUI reflects the weighted average of forecast errors, making it largely independent of growth levels.

cial Conditions Index (FCI) proposed by Carrillo and García (2021).¹³ These authors analyze the response of several economic activity indicators to a shock in financial conditions. According to their estimates, economic activity declines in a U-shape manner for about twelve months after a shock to financial conditions. Production and consumption responses are of similar magnitudes, while investment falls to a greater extent.

Figure 2
*IGAE (yoy) growth and financial
conditions index for Mexico
April 94 - December 21*



Notes: To facilitate comparison, both series are standardized.

Source: Authors' elaboration.

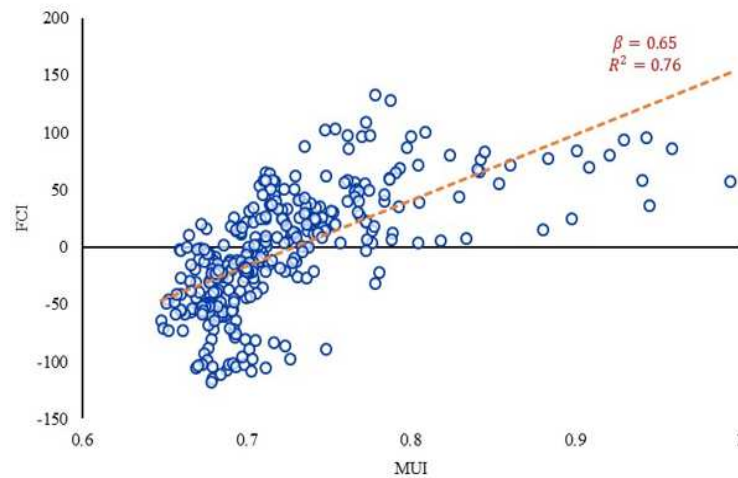
The FCI and economic activity in Mexico, measured as the annual growth rate of IGAE, also show a strong negative correlation (see Figure 2).¹⁴ In addition, there is a significant positive relationship between the MUI and the FCI (see Figure 3).

¹³ These authors calculate their FCI following a methodology similar to the one proposed by Hatzius and Stehn (2018) and Koop and Korobilis (2014). In particular, they estimate a dynamic factor model using a Kalman filter adjusted for unobserved values and an unbalanced sample for all variables included. These variables are divided into seven categories: currencies, stocks, debt, uncertainty, country risk, commodity prices, and economic activity.

¹⁴ Banxico (2019, 2020a) present evidence about the impact of financial conditions on the distribution of GDP growth in Mexico in a framework similar to the one proposed by Adrian *et al.* (2019), using in both analyses a FCI provided by the Bank of Mexico different from the one proposed by Carrillo and García (2021).

In this regard, it may be possible that a tightening of financial conditions will be simply a reflection of an increase in the level of uncertainty. For instance, in environments of greater uncertainty, firms and households could face more restrictive credit and liquidity conditions associated with a greater aversion of banks to grant loans and higher costs of credit access.¹⁵

Figure 3
*Financial conditions index and
macroeconomic uncertainty index,
for Mexico*



Source: Authors' elaboration.

To further study the relationship between growth and uncertainty in Mexico, we start by estimating a VAR similar to Bonciani and Jason (2019), analyzing impulse-response functions (IRF), historical decomposition (HD), and forecast error variance decomposition (FEVD).¹⁶ Shocks are identified using a Cholesky identification. The

¹⁵ A Granger causality test with three lags reveals a significant relationship. MUI is found to cause FCI (p-value = 0.00), while the influence of FCI on MUI is not as strong (p-value = 0.07) at the 0.05 significance level.

¹⁶ In a related analysis, Banxico (2019) also analyzes the effect of uncertainty on consumption and investment in Mexico by estimating two VAR models, one for consumption and the other for investment.

VAR includes 10 endogenous variables: The United States industrial production index (IPUS), MUI, FCI, nominal exchange rate (NER), IGAE, private consumption (PCI), gross fixed investment (GFI), consumer price index (INPC) and government securities rate (CETES).¹⁷ The ordering of endogenous variables implies that IPUS is assumed to be the most exogenous variable, as Mexico's economy depends on the United States activity. The MUI responds contemporaneously by shocks to the IPUS but not to other variables.¹⁸ This identification is in line with recent literature showing that macroeconomic uncertainty can be considered an exogenous business cycle driver (for more details, see Bonciani and Jason, 2019). The VAR is estimated with three lags¹⁹ to identify the impact of uncertainty shocks on the endogenous variables in the short term, and IRF/FEVD are calculated for a 36-month horizon.

The impulse-response functions between IGAE growth, MUI, and the rest of the endogenous variables provide evidence of a negative, statistically significant, effect of uncertainty on economic growth.²⁰ A one standard deviation (sd) shock in uncertainty reduces IGAE growth

¹⁷ Regarding the sources of the data, IGAE, PCI, GFI and INPC are taken from the Economic Information Bank from INEGI; NER and CETES are taken from the Economic Information System from the Bank of Mexico; the IPUS is taken from the Federal Reserve Bank of St. Louis. Finally, the MUI and the FCI are provided by the Bank of Mexico. All variables are in percentage annual growth rate variation, with the exception of MUI, FCI, and CETES and seasonal adjusted with the exception of NER, INPC and CETES.

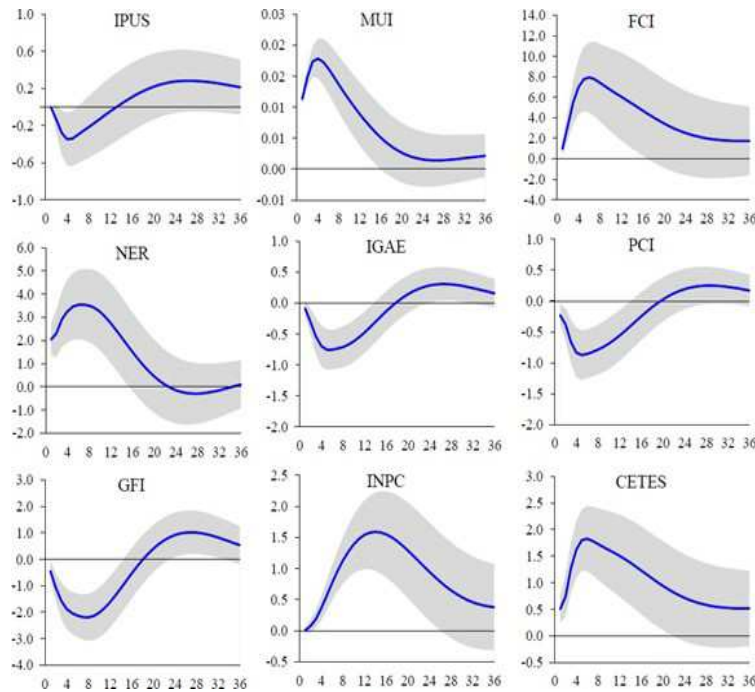
¹⁸ With the order of identification of the variables, we assume the MUI is relatively more exogenous than the FCI in the sense that shocks to the MUI affect the FCI contemporaneously, but not the other way around. Clearly, this assumption is restrictive, and it could be suggested that an order in which it is assumed that financial conditions are more exogenous than uncertainty is more appropriate. We carried out a VAR specification in which the Cholesky ordering for the FCI and the MUI is interchanged, reaching similar results. Due to the length of the article we chose not to include this exercise, but it is available upon request.

¹⁹ The number of lags was determined through the Akaike information criterion. Other information criteria also suggest the same number of lags.

²⁰ We did not find an effect significantly different from zero of the FCI on IGAE annual growth. This result does not necessarily imply that the FCI is not an important determinant of growth, since the relevant information that this contains could be more useful in explaining other moments of the growth distribution different from the mean.

by over 0.5 sd, with the effect lasting about 11 months. Figure 4 shows the IRF of endogenous variables after an MUI shock, with responses aligning with expectations: the exchange rate rises, private consumption falls in proportion to IGAE, and gross fixed investment declines more sharply.²¹

Figure 4
Response to Cholesky one s.d. (d.f. adjusted)
innovations to MUI
April 94 - December 21

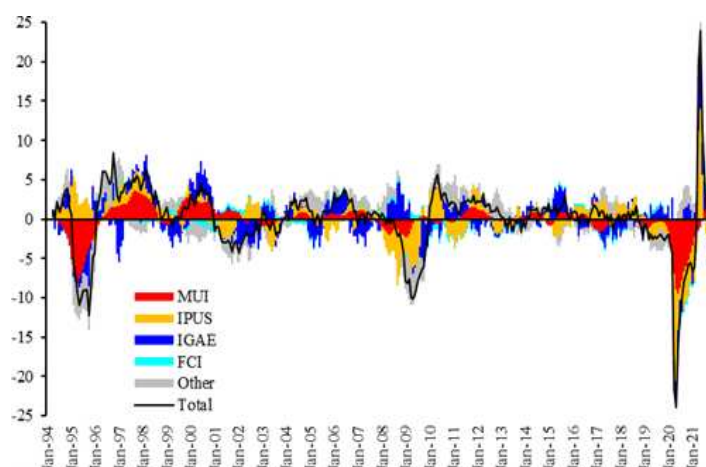


Source: Authors' elaboration.

²¹ Including the COVID-19 period in the sample may bias the VAR estimates. To assess this, we conducted an additional analysis using data up to December 2019. Results were similar, but the uncertainty shocks impact was smaller, indicating overestimation when including the pandemic period. This is reasonable given the size of the shocks during that time. Additionally, while the pre-2001 period could bias results due to changes in the exchange rate and monetary policy regimes, we include it since our focus is on uncertainty shocks, not those related to exchange rates or policy. Including this subsample also provides more flexibility, given the number of parameters estimated.

The HDS provide evidence about the incidence of each shock on the variation of economic activity. As can be seen in Figure 5, during the period of the COVID-19 pandemic between April 2020 and December 2020, more than half of the negative variation of the IGAE can be attributed to the level of uncertainty in Mexico, highlighting the relatively low contribution of financial conditions.²² Compared to other deep crises, such as those of 1994 and 2008, it is interesting to note that although uncertainty explains a significant proportion of the fall in production, it is relatively less important than other shocks.

Figure 5
Historical decomposition of IGAE (yoy) growth
January 94 - December 21



Notes: Other is the aggregated contribution of NER, PCI, GFI, INPC, and CETES.

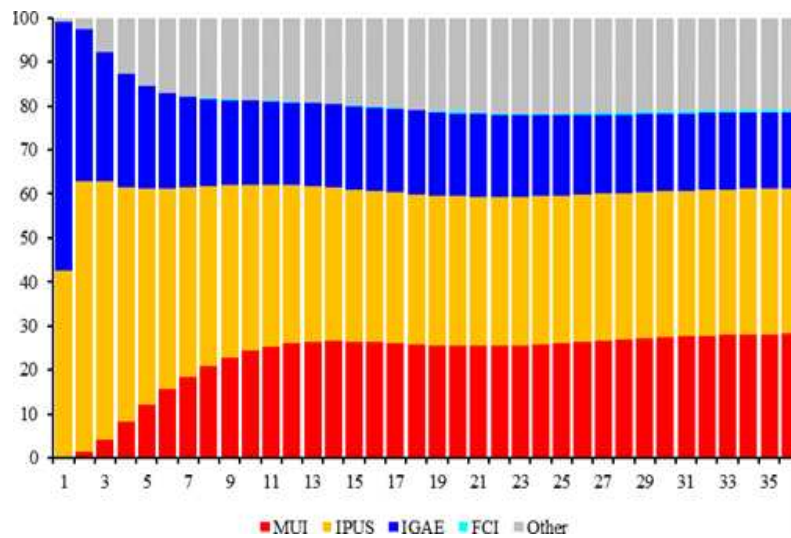
Source: Authors' elaboration.

Finally, according to FEVD estimates (see Figure 6), after 36 months, around 25% of IGAE growth is explained by uncertainty shocks, while production shocks explain about 43%. In contrast, FCI shocks explain only a small proportion of the variance of the growth forecast error. Bonciani and Jason (2019) present similar results to

²² This result also holds under Cholesky's alternative identification, where uncertainty is considered relatively more exogenous than the FCI.

ours for the United States economy. Likewise, in the case of Mexico, Banxico (2019) analyzes the impact of MUI on investment and consumption, finding similar results.²³

Figure 6
*Forecast error variance decomposition of
IGAE (yoy) growth*



Notes: Other is the aggregated contribution of NER, PCI, GFI, INPC, and CETES.

Source: Authors' elaboration.

3. Growth at risk and macroeconomic uncertainty in Mexico

3.1 *Growth at risk methodology*

The previous section presented evidence that annual IGAE growth in Mexico responds negatively and significantly to MUI shocks. While these results are robust to the inclusion of other growth determinants like the FCI (Carrillo and García, 2021), point estimates can ignore the

²³ However, unlike previous approaches, we control for financial conditions in Mexico in our analysis to identify the effect of uncertainty on the dynamics of economic activity.

negative biases in growth expectations. Therefore, estimating the entire conditional distribution of growth is more suitable for evaluating uncertainty's impact (Adrian *et al.*, 2019).

To capture the full conditional distribution of economic growth and assess the role of uncertainty, we apply the two-stage methodology proposed by Adrian *et al.* (2019). The first stage uses quantile regression to model the annual growth rate of IGAE against a set of control variables. In the second stage, the parameters of a *skewed-t* distribution are estimated using the quantile regression results. This approach enables a comprehensive analysis of the determinants' marginal effects along the entire conditional distribution.

3.1.1 Step 1: Quantile regression

The quantile function, $F^{-1}(\tau)$, maps a given quantile τ to its corresponding value x such that:

$$F^{-1}(\tau) = \inf \{x \in \mathbb{R}: \tau \leq F(x)\} \quad (1)$$

Assuming the quantile function of the dependent variable y_{t+h} , conditional on x_t , is linear in parameters, $Q_{y_{t+h}|x_t}(\tau|x_t) = x_t\beta_\tau$, the coefficients β_τ are obtained by minimizing the weighted absolute value of the residuals:

$$\hat{\beta}_\tau = \arg \min_{\beta} \sum_{t=1}^{T-h} (\tau \cdot 1(y_{t+h} \geq x_t\beta) \cdot |y_{t+h} - x_t\beta| + (1 - \tau) \cdot 1(y_{t+h} < x_t\beta) \cdot |y_{t+h} - x_t\beta|) \quad (2)$$

The fitted value of y_{t+h} , conditional on x_t , is then expressed as:

$$\hat{Q}_{y_{t+h}}(\tau|x_t) = x_t\hat{\beta}_\tau \quad (3)$$

Quantile regression estimates are particularly useful for capturing the asymmetric effects of uncertainty and other determinants along the entire growth distribution. They help identify whether these effects are concentrated in specific regions of the distribution, such as the left tail, which indicates a higher risk of low growth.

3.1.2 Step 2: Continuous distribution estimation

The second stage involves estimating the parameters ψ_{t+h} of a continuous distribution, such as the *skewed-t*, by minimizing the distance

between the quantile regression fitted values and the inverse of the cumulative distribution function:

$$\left\{ \hat{\psi}_{t+h} \right\} = \arg \min_{\psi} \sum_{\tau} \left(\hat{Q}_{y_{t+h}|x_t}(\tau|x_t) - F^{-1}(\tau; \psi) \right)^2 \quad (4)$$

The *skewed-t* distribution, with parameters $(\mu, \sigma, \alpha, \nu)$ is defined as:

$$f(y; \mu, \sigma, \alpha, \nu) = \frac{2}{\sigma} s\left(\frac{y - \mu}{\sigma}; \nu\right) S\left(\alpha \frac{y - \mu}{\sigma} \sqrt{\frac{\nu + 1}{\nu + (y - \mu/\sigma)^2}}; \nu + 1\right) \quad (5)$$

where $s(\cdot)$ and $S(\cdot)$ represent the density and cumulative distribution functions of a t distribution, respectively. This flexible distribution captures both asymmetry and kurtosis, making it suitable for modeling the conditional distribution of expected growth under uncertainty.

For each period, the four parameters of the *skewed-t* distribution are estimated using maximum likelihood. These parameters describe a continuous distribution of expected IGAE growth, allowing for a nuanced analysis of the effects of uncertainty.

Using this method, we derive two distinct distributions: one conditional on MUI and one without MUI. This enables us to assess the distinct impact of uncertainty by comparing the two distributions.

3.2 Data and benchmark model

The GaR estimates utilize the same monthly dataset employed in the previous VAR analysis. The IGAE is used as the primary measure of economic activity in Mexico, while control variables include lags of IGAE, IPUS, inflation (INF), nominal exchange rate (NER), and CETES rates. The impact of MUI on various quantiles of IGAE growth is assessed using the following benchmark model:

$$y_{t+h}^{\tau} = \beta_0^{\tau} + \beta_1^{\tau} y_t + \beta_2^{\tau} INF_t + \beta_3^{\tau} NER_t + \beta_4^{\tau} CETES_t + \beta_5^{\tau} IPUS_t + \beta_6^{\tau} MUI_t + \varepsilon_t^{\tau} \quad (6)$$

where y_t represents the annual growth rate of IGAE, INF_t denotes annual inflation, $IPUS_t$ is the annual growth rate of industrial production in the United States, and MUI_t is the macroeconomic uncertainty index with a one-month forecast horizon. The superscript h indicates the forecast horizon, and τ denotes the quantile being estimated, which ranges from 0.1 to 0.9 at 0.05 intervals. For example, y_{t+3}^{25} refers to the 25th quantile of IGAE growth distribution with a three-month forecast horizon.

3.3 Impact of uncertainty on expected growth distribution in Mexico

Figure 7 presents the estimated coefficients of equation 6 for each quantile of the annual IGAE growth distribution at a three-month forecast horizon.²⁴ The results indicate that uncertainty exerts a negative and significant effect on the left tail of the distribution. In contrast, its impact on the right tail is not statistically significant. This suggests that macroeconomic uncertainty primarily increases downside risks of expected growth.

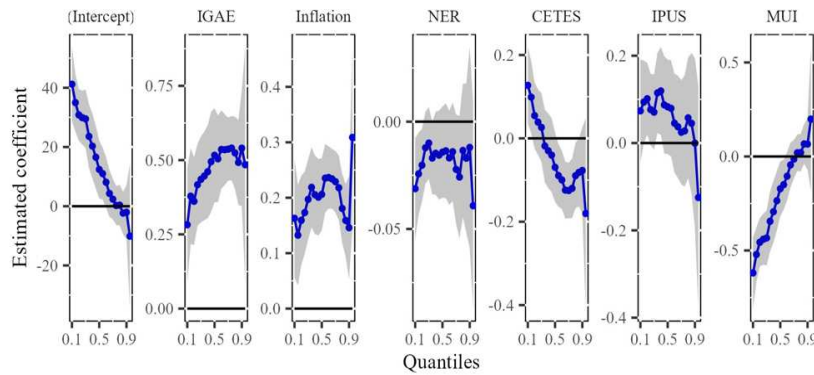
Specifically, the effect of MUI on the 10th quantile is -0.62 percentage points (pp), significantly higher than its impact on the median (-0.17 pp) (Table 1).

Further analysis shows that the negative effect of MUI on growth is more pronounced for lower quantiles, indicating that uncertainty has an asymmetric impact on the growth distribution. Figure 9 illustrates the negative correlation between the interquartile range (a measure of dispersion) and the median, highlighting that increased uncertainty leads to a wider dispersion and a lower median of expected growth. This result suggests a shift in the distribution to the left, indicating higher probability mass associated with low levels of growth, similar to findings in other economies (e.g., the United States and China) (Gu *et al.*, 2021; Jovanovic and Ma, 2020).

Figure 8 presents the estimated coefficients of the Benchmark Model for the MUI across various forecast horizons, h . The results indicate that, for forecast horizons ranging from one to 12 months, the marginal effect of uncertainty on expected IGAE growth is negative and increases in absolute magnitude for the lower quantiles of the distribution. Notably, this finding is consistent with the results of Jovanovic and Ma (2020) for the case of the United States.

²⁴ Results hold for forecast horizons from one to 15 months.

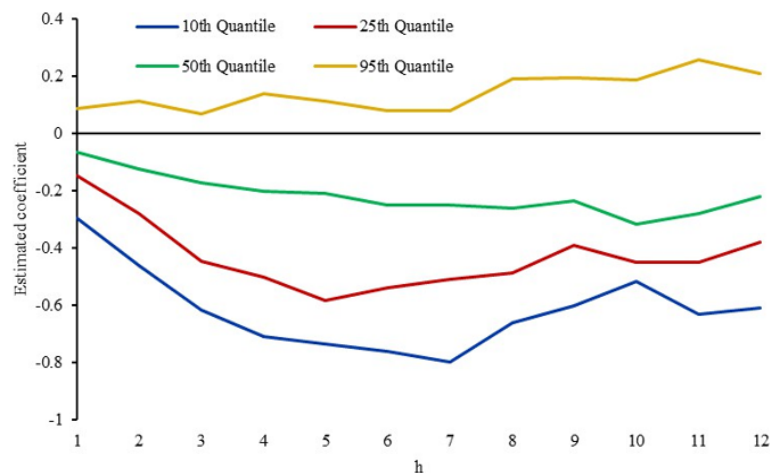
Figure 7
*Quantile regression coefficients for
the benchmark model ($h = 3$ months)*



Notes: Blue lines represent coefficients for each quantile, and the grey band depicts 90% confidence intervals.

Source: Authors' elaboration.

Figure 8
Estimated coefficients for MUI



Source: Authors' elaboration.

Table 1
Estimated coefficients of benchmark model
($h = 3$ months; Sample: April 94 - December 21)

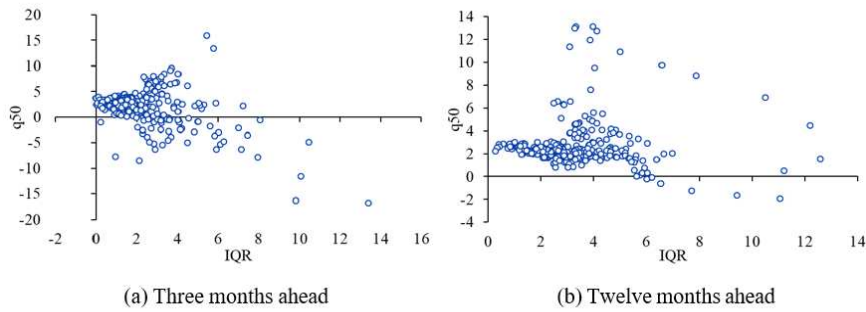
<i>(a) 10th Quantile</i>				<i>(b) 25th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	41.08 ***	28.4	53.77	Intercept	30.42 ***	21.5	39.32
IGAE	0.29 ***	0.11	0.48	IGAE	0.42 ***	0.33	0.56
Inflation	0.17 **	0.06	0.28	Inflation	0.17 ***	0.10	0.24
NER	-0.03 **	-0.06	-0.01	NER	0.00	-0.01	0.01
CETES	0.13 **	0.03	0.22	CETES	0.04	0.00	0.12
IPUS	0.07	-0.05	0.19	IPUS	0.07	0.00	0.15
MUI	-0.62 ***	-0.81	-0.43	MUI	-0.45 ***	-0.58	-0.32
<i>(c) 50th Quantile</i>				<i>(d) 90th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	12.57 **	2.76	22.37	Intercept	-2.03	-11.19	7.12
IGAE	0.52 ***	0.36	0.68	IGAE	0.54 ***	0.34	0.74
Inflation	0.20 ***	0.14	0.27	Inflation	0.15 ***	0.06	0.24
NER	-0.02	-0.04	0.01	NER	-0.01	-0.06	0.04
CETES	-0.07	-0.15	0.02	CETES	0.03	-0.05	0.11
IPUS	0.08	0.00	0.16	IPUS	0.07	0.00	0.14
MUI	-0.17 *	-0.32	-0.03	MUI	0.08	-0.03	0.19

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

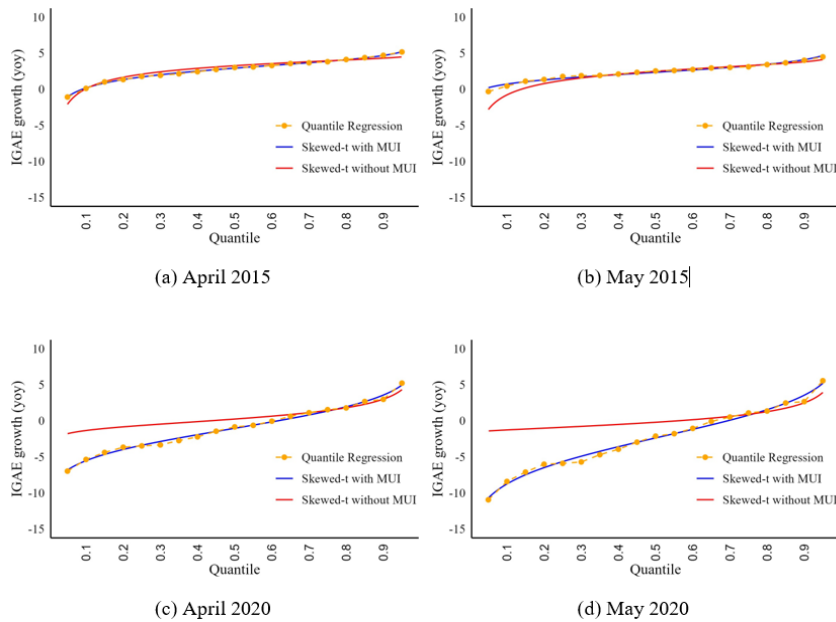
To further examine uncertainty's role, we compare the cumulative distribution of IGAE growth conditional and unconditional on MUI for specific periods. Figure 10 shows that during low-uncertainty periods (e.g., April 2015), the conditional distribution closely aligns with the unconditional one. Conversely, in high-uncertainty periods (e.g., April 2020), the conditional distribution shifts significantly leftward, indicating a substantial increase in downside risks.

Figure 9
*Median and interquartile range scatterplot
of quantile regression*



Source: Authors' elaboration.

Figure 10
*Quantile regression and the skewed-t distribution
($h = \text{three months}$)*



Source: Authors' elaboration.

3.4 Relative probability gain and entropy

The previous section showed that under high uncertainty, the conditional distribution *skewed- t with MUI* assigns a greater probability mass to low growth levels compared to the *skewed- t without MUI* for a three-month horizon.²⁵ This result suggests that conditioning the distribution of expected growth on the MUI results in a higher relative probability gain (RPG) for low growth predictions when uncertainty rises.

The RPG for each period t , and a given growth threshold g_t , is defined as the difference between the cumulative conditional and unconditional distributions:

$$RPG_t^h(g_t) = F_{IGAE}^h(g_t; X_{t-h}, MUI_{t-h}) - G_{IGAE}^h(g_t; X_{t-h}) \quad (7)$$

Where $F_{IGAE}^h(g_t; X_{t-h}, MUI_{t-h})$ is the conditional cumulative distribution with MUI and $G_{IGAE}^h(g_t; X_{t-h})$ is the unconditional cumulative distribution without MUI. The control variables X_{t-h} include IGAE, IPUS, and inflation.

We calculate the average RPG for negative growth ($g_t=0$) during the 1994 crisis, the 2008 financial crisis, and the COVID-19 pandemic.²⁶ Figure 11 shows that RPG is significant for these crisis periods, with no substantial RPG observed for non-crisis periods. Additionally, RPG and MUI exhibit a strong positive correlation, indicating that higher uncertainty is associated with a higher probability that a negative growth level will be realized.

Relative entropy provides an alternative measure of growth risk by quantifying the “extra” probability mass that the conditional distribution assigns to extreme outcomes compared to the unconditional distribution (Adrian *et al.* 2019). When the lower relative entropy (LRE) is high, the conditional density assigns a greater probability to low growth outcomes. Conversely, high upper relative entropy (URE) indicates a greater probability for high growth outcomes. Formally, LRE and URE for forecast horizon h in period t are defined as:

$$E_{t,h}^l = - \int_{-\infty}^{\hat{F}_h^{-1}(0.5|x_t, MUI_t)} \log \left(\frac{\hat{g}_h(y|x_t)}{\hat{f}_h(y|x_t, MUI_t)} \right) \hat{f}_h(y|x_t, MUI_t) dy \quad (8)$$

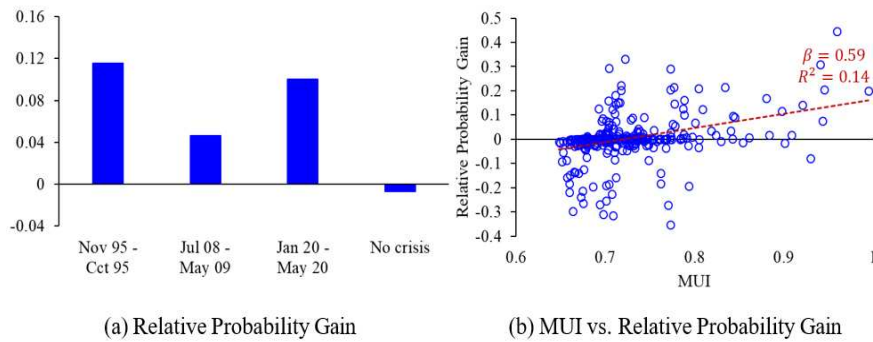
²⁵ This pattern holds for one, six, and twelve-month horizons as well.

²⁶ For each crisis, we consider periods of sustained IGAE decline: Nov-1994 to Oct-1995, Jul-2008 to May-2009, and Jan-2020 to May-2020.

$$E^s_{t,h} = - \int_{\hat{F}_h^{-1}(0.5|x_t, MUI_t)}^{\infty} \log \left(\frac{\hat{g}_h(y|x_t)}{\hat{f}_h(y|x_t, MUI_t)} \right) \hat{f}_h(y|x_t, MUI_t) dy \quad (9)$$

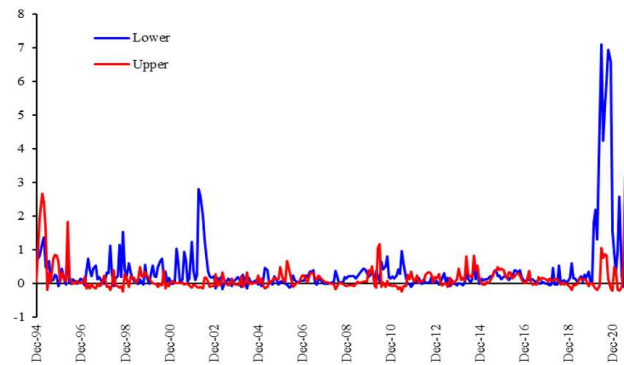
Where $\hat{g}_h(y|x_t)$ is the unconditional density, $\hat{f}_h(y|x_t, MUI_t)$ is the conditional density with MUI, and $\hat{F}_h^{-1}(0.5|x_t, MUI_t)$ is the median of the conditional density.

Figure 11
Relative probability gain and MUI



Source: Authors' elaboration.

Figure 12
Relative entropy index



Source: Authors' elaboration.

Figure 12 shows elevated downside risks (high LRE) during the 1994 crisis and the COVID-19 pandemic. Interestingly, upside risks (high URE) were also present during the 1994 crisis, likely due to the high dispersion in growth expectations at the time. However, no significant downside risks were detected during the 2008 financial crisis, likely due to Mexico's stronger macroeconomic foundations and financial regulation at that time.

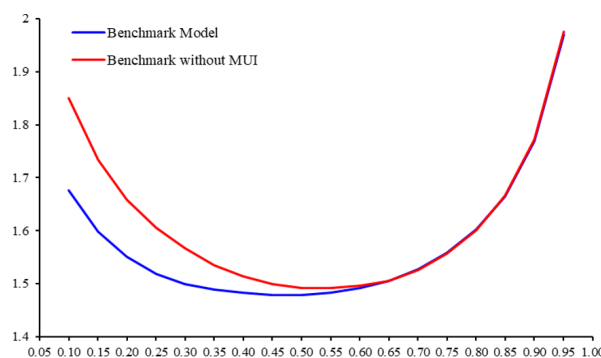
3.5 Robustness analysis

In this section, we present some robustness tests of the reference model that consist of some diagnostic tests and additional exercises with alternative subsamples that do not include periods of high uncertainty, the use of alternative measures of economic activity, the inclusion in the base model of measures of the state of financial conditions prevailing in Mexico and other alternative measures of uncertainty to assess the validity of the impact of the MUI on the distribution of expected growth.

3.5.1 Diagnostic tests of the model

Diagnostic tests confirm the robustness of our model. Ramsey's RESET test shows no specification errors, and the slope equality test indicates no significant asymmetries in the residuals (Table 2). Furthermore, the Akaike Information Criterion (AIC) suggests that the model including MUI outperforms the one without it for lower quantiles, underscoring the relevance of uncertainty for explaining lower growth outcomes (Figure 13).

Figure 13
Akaike information criterion (AIC)



Source: Authors' elaboration.

Table 2
Diagnostic tests

<i>Test</i>	<i>Statistic</i>	<i>P value</i>	<i>Decision</i>
Ramsey RESET Test	0.18	0.67	No specification error
Symmetric Quantiles Test	8.99	0.83	Quantiles errors are symmetric
Hypothesis Test	2.55	0.03	10 th and 50 th quantiles are statistically different

Source: Authors' elaboration.

3.5.2 Restricted samples without episodes of high uncertainty

We re-estimate the benchmark model using two restricted samples: from April 1994 to December 2019 (excluding the COVID-19 pandemic) and from January 2001 to December 2019 (excluding the 1994 crisis and COVID-19). Table 3 shows that the effect of MUI remains negative and significant in the left tail, although the magnitude is reduced compared to the full sample. This suggests that MUI remains a reliable indicator for predicting economic growth on the left tail of the distribution even in relatively stable periods.

Table 3
Estimated coefficients for MUI of benchmark model
(h = 3 months, subsamples)

<i>(a) Sub-sample: Apr-94 - Dec-19</i>				<i>(b) Sub-sample: Jan-01 - Dec-19</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.45 ***	-0.54	-0.36	10 th Quantile	-0.22 **	-0.47	-0.08
25 th Quantile	-0.36 ***	-0.48	-0.27	25 th Quantile	-0.21 ***	-0.27	-0.12
50 th Quantile	-0.24 ***	-0.40	-0.15	50 th Quantile	-0.14 **	-0.22	-0.09
90 th Quantile	0.01	-0.19	0.22	90 th Quantile	0.20	-0.05	0.48

Notes: *p<0.1, **p<0.05, ***p<0.01. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

3.6 Alternative measures of economic activity

We further test the robustness of our results using alternative indicators of economic activity, including private consumption, gross fixed investment, industrial production, and tertiary sector activities. We also analyze labor market indicators such as formal employment and the employment rate. All variables are expressed as annual growth rates, except for the employment rate, which is included in levels and defined as the share of employment over the labor force. Our results, presented in Table 4, remain consistent across these indicators, showing that uncertainty has a pervasive impact on growth distribution across various economic sectors.

Table 4
*Estimated coefficients for MUI of benchmark model for
 alternative measures of economic activity
 (h = 3 months; sample: April 94 - December 21)*

<i>(a) Consumption</i>				<i>(b) Investment</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.67 ***	-0.92	-0.42	10 th Quantile	-1.71 ***	-2.48	-0.95
25 th Quantile	-0.36 ***	-0.52	-0.20	25 th Quantile	-0.95 ***	-1.38	-0.52
50 th Quantile	-0.18 ***	-0.29	-0.08	50 th Quantile	-0.38	-0.83	0.08
90 th Quantile	0.13	-0.01	0.28	90 th Quantile	0.05	-0.27	0.37
<i>(c) Industrial production</i>				<i>(d) Services</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.66 ***	-0.85	-0.47	10 th Quantile	-0.64 ***	-0.84	-0.43
25 th Quantile	-0.57 ***	-0.79	-0.35	25 th Quantile	-0.39 ***	-0.51	-0.26
50 th Quantile	-0.19	-0.41	0.03	50 th Quantile	-0.13	-0.27	0.00
90 th Quantile	0.13	-0.03	0.30	90 th Quantile	0.02	-0.12	0.16
<i>(e) Employment level</i>				<i>(f) Employment rate</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.19 ***	-0.30	-0.08	10 th Quantile	-0.07 ***	-0.09	-0.05
25 th Quantile	-0.18 ***	-0.27	-0.09	25 th Quantile	-0.07 ***	-0.10	-0.04
50 th Quantile	-0.07 *	-0.14	-0.01	50 th Quantile	-0.04 ***	-0.06	-0.02
90 th Quantile	-0.02	-0.08	0.05	90 th Quantile	-0.01	-0.03	0.00

Notes: *p<0.1, **p<0.05, ***p<0.01. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

Table 5
Estimated coefficients of model with GDP (yoy) growth
(h = 1 quarter) 94-Q2 - 21-Q4

<i>(a) 10th Quantile</i>				<i>(b) 25th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	37.32 **	10.0	64.7	Intercept	27.37 ***	17.8	37.0
Inflation	0.27 *	0.0	0.5	Inflation	0.21 ***	0.1	0.3
IPUS	0.08	-0.1	0.3	IPUS	0.08	0.0	0.2
MUI	-0.56 **	-1.0	-0.1	MUI	-0.41 ***	-0.5	-0.3
GDP	0.34 *	0.0	0.7	GDP	0.56 ***	0.3	0.8
<i>(c) 50th Quantile</i>				<i>(d) 90th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	26.69 ***	10.4	43.0	Intercept	3.56	-7.5	14.6
Inflation	0.23 ***	0.1	0.4	Inflation	0.12 ***	0.0	0.2
IPUS	0.01	-0.1	0.2	IPUS	-0.18	-0.5	0.1
MUI	-0.39 **	-0.6	-0.1	MUI	0.32	-0.4	1.0
GDP	0.56 ***	0.3	0.8	GDP	0.69 ***	0.4	1.0

Notes: *p<0.1, **p<0.05, ***p<0.01. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

Investment growth appears particularly sensitive to uncertainty. The estimated marginal effect of MUI on investment growth is -1.71 pp for the 10th quantile, compared to a range of -0.07 to -0.67 pp for other indicators, highlighting the stronger influence of uncertainty on investment growth expectations.

Finally, we conduct robustness checks using quarterly GDP data instead of monthly IGAE. These results, shown in Table 5, are consistent with those using IGAE, indicating that uncertainty increases both the dispersion and the negative bias of expected GDP growth. This suggests that our findings are not driven by data frequency or by potential endogeneity issues related to the use of IGAE in the construction of MUI.

Overall, the robustness analyses confirm that MUI remains a strong predictor of negative growth expectations in Mexico, even when using alternative measures of economic activity.

3.7 *Alternative measures of uncertainty and financial conditions*

Our findings confirm the significant impact of uncertainty on the distribution of expected IGAE growth in Mexico. To validate these results, we incorporate additional measures related to financial conditions, latent risk, and uncertainty, which might be correlated with the MUI. For instance, some evidence suggests that financial conditions may influence economic performance in Mexico. For instance, Carrillo and García (2021) found that negative financial shocks adversely affect GDP, consumption, and investment. To test this hypothesis, we include an additional control in our benchmark model in the following way:

$$y_{t+h}^{\tau} = \beta_0^{\tau} + \beta_1^{\tau} y_t + \beta_2^{\tau} INF_t + \beta_3^{\tau} NER_t + \beta_4^{\tau} CETES_t + \beta_5^{\tau} IPUS_t + \beta_6^{\tau} MUI_t + \theta_{\tau} AD_t + \varepsilon_t^{\tau} \quad (10)$$

Where AD_t can be either a measure of local or external financial conditions, other measures of uncertainty and risk as specified below.

Results indicate that once MUI is included, FCI no longer has a significant effect on the distribution of expected IGAE growth. The MUI coefficients remain negative and statistically significant for the left tail of the distribution, consistent with the benchmark model (Table 6). This suggests that uncertainty is a key determinant of growth, even after controlling for financial conditions.

Next, we include additional indicators related to financial conditions and uncertainty, such as the United States NFCI and Economic Policy Uncertainty (EPU) indices for Mexico, the United States, and globally. These additions do not alter our conclusions, indicating that MUI captures unique information about uncertainty's impact on growth expectations that other measures do not. This contrasts with Gu *et al.* (2021) for China, where EPU remained significant even when other uncertainty indices were included.

We also test other traditional risk indicators, such as the real exchange rate and the risk premium (defined as the difference between the United States federal funds rate and CETES rate). Results show that exchange rate depreciation, both nominal and real, has a significant negative impact on the left tail of the distribution, suggesting that MUI and exchange rate variations may be complementary in explaining negative biases on expected growth in Mexico.

Table 6

*Estimated coefficients for MUI of benchmark model for
alternatives measures of uncertainty and financial conditions
($h = 3$ months; Sample: April 94 - December 21)*

<i>(a) NFCI</i>				<i>(b) EPU</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.63 ***	-0.88	-0.37	10 th Quantile	-0.57 ***	-0.79	-0.35
25 th Quantile	-0.44 ***	-0.59	-0.28	25 th Quantile	-0.32 ***	-0.47	-0.18
50 th Quantile	-0.17 **	-0.30	-0.05	50 th Quantile	-0.08	-0.21	0.06
90 th Quantile	0.06	-0.08	0.19	90 th Quantile	0.07	-0.14	0.28
<i>(c) EPU US</i>				<i>(d) EPU Global</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.61 ***	-0.91	-0.31	10 th Quantile	-0.57 ***	-0.79	-0.35
25 th Quantile	-0.32 ***	-0.48	-0.15	25 th Quantile	-0.32 ***	-0.47	-0.17
50 th Quantile	-0.06	-0.20	0.08	50 th Quantile	-0.09	-0.22	0.05
90 th Quantile	0.07	-0.16	0.31	90 th Quantile	0.07	-0.15	0.28
<i>(e) RER</i>				<i>(f) Risk premium</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
10 th Quantile	-0.62 ***	-0.8	-0.44	10 th Quantile	-0.61 ***	-0.8	-0.42
25 th Quantile	-0.44 ***	-0.57	-0.31	25 th Quantile	-0.44 ***	-0.58	-0.30
50 th Quantile	-0.17 *	-0.32	-0.02	50 th Quantile	-0.17 *	-0.32	-0.02
90 th Quantile	0.07	-0.01	0.21	90 th Quantile	0.04	-0.10	0.18

Notes: *p<0.1, **p<0.05, ***p<0.01. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

3.8 Crossing forecasts

In quantile regression forecasting, out-of-sample forecasts may cross due to several reasons related to the properties of the estimated quantile functions and the constraints applied during the estimation. Forecast crossing refers to a situation where the predicted values for lower quantiles are higher than those for upper quantiles. This can occur because:

- Lack of constraints: When fitting quantile regressions independently for different quantile levels, the resulting quantile estimates might cross if no explicit non-crossing constraints are applied. This is because each quantile regression is estimated independently, and there is no guarantee that the estimated functions will be ordered appropriately across quantiles.
- Estimation error and model specification: Forecast crossing can also arise due to estimation error or misspecification of the model. If the underlying data-generating process is not captured well by the model, the resulting quantile estimates may exhibit crossing.
- Finite sample issues: In finite samples, the estimated quantiles are subject to sampling variability, which may result in crossing even if the true quantile functions are non-crossing.

Quantile crossings are generally considered invalid in quantile regression because they violate a fundamental principle: the probability of a lower quantile should always be less than or equal to that of a higher quantile. When this condition is not met, it introduces inconsistency, which can lead to incorrect probabilistic interpretations.

The crossing forecasting problem does not seem to be a major issue in our analysis. An inspection of the fitted values of our model for different quantiles shows minimal crossings relative to the full sample.²⁷ Also, the inclusion of several variables that are important for predicting growth from an economic point of view and the tests applied to the model suggest that there are no specification problems. However, based on Bondell *et al.* (2010) we estimate our based model by imposing non-crossing conditions. Table 7 shows results of this alternative estimation. Although the estimated coefficients are lower in absolute value than those of our base model, they are in line with our main results and show a greater vulnerability of the expected growth of the IGAE in the left tail of its distribution.

²⁷ Quantile crossings occurred in only 3 of 333 periods, representing less than 1% of the sample. As the analysis is in-sample and these specific periods are not relevant to the core findings, the results remain valid despite the minor crossings.

Table 7
*Estimated coefficients for MUI of benchmark model with
 non-crossing quantiles restrictions
 ($h = 3$ months; Sample: April 94 - December 21)*

<i>(a) 10th Quantile</i>				<i>(b) 25th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	-11.82 ***	-15.62	-8.01	Intercept	-11.81 ***	-15.1	-8.57
IGAE	0.50 ***	0.32	0.68	IGAE	0.52 ***	0.40	0.67
Inflation	0.22 **	0.06	0.39	Inflation	0.22 ***	0.14	0.31
NER	-0.01	-0.05	0.04	NER	-0.02	-0.07	0.03
CETES	-0.06	-0.26	0.14	CETES	-0.06	-0.2	0.07
IPUS	0.11 **	0.02	0.2	IPUS	0.11 **	0.00	0.19
MUI	-0.36 ***	-0.54	-0.19	MUI	-0.30 ***	-0.47	-0.12
<i>(c) 50th Quantile</i>				<i>(d) 90th Quantile</i>			
	<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>		<i>Coef.</i>	<i>Lower</i>	<i>Upper</i>
Intercept	-11.01 ***	-14.07	-7.95	Intercept	-8.52 ***	-12.13	-4.9
IGAE	0.51 ***	0.37	0.66	IGAE	0.49 ***	0.34	0.67
Inflation	0.21 ***	0.15	0.27	Inflation	0.16 ***	0.07	0.24
NER	-0.01	-0.04	0.01	NER	-0.02	-0.05	0.01
CETES	-0.04	-0.12	0.03	CETES	-0.04	-0.18	0.10
IPUS	0.08	-0.01	0.18	IPUS	0.07	-0.04	0.19
MUI	-0.17	-0.36	0.02	MUI	0.05	-0.07	0.16

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Lower and Upper indicate the 90% confidence interval.

Source: Authors' elaboration.

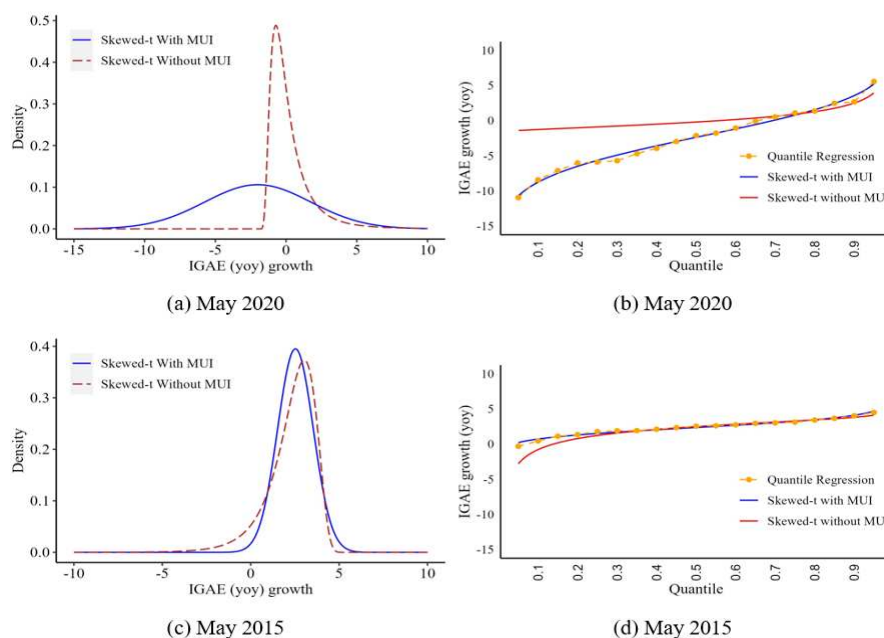
4. Analysis of uncertainty during the COVID-19 pandemic

In this section, we present a brief analysis of uncertainty and its effects on growth during the COVID-19 pandemic, since our models estimates are particularly relevant for analyzing recessions, as they reveal how uncertainty exacerbates the negative bias in growth expectations.

As shown in Figure 1, MUI estimates identify the pandemic as a period of extremely high uncertainty levels. During this time, actual

economic activity aligned more closely with the lower quantiles of the expected growth distribution, rather than central moments like the median. This suggests that uncertainty magnified the negative bias in growth projections, making central estimates appear overly optimistic. In particular, since the beginning of 2020, in response to the COVID-19 health crisis, health authorities implemented social distancing measures and the closure of non-essential activities, emulating the responses implemented in other countries. These actions translated into shocks to the economy, both on the demand and supply sides.

Figure 14
Probability density function and cumulative distribution function



Notes: Distributions are estimated from the benchmark model. Cumulative density function plots (b and d) have the quantile on the x-axis. Therefore, it should be interpreted as the probability of having a y-o-y growth less than or equal to the value on the y-axis.

Source: Authors' elaboration.

The estimated probability distributions of expected growth for May 2020, a period in which the greatest historical drop in economic

activity was recorded and also the greatest increase in uncertainty registered through the MUI (Figure 14), show an increase in the negative bias of growth expectations (Figure 11). Likewise, our estimates of relative entropy show a very important increase in the vulnerability of growth, greater magnitude than those registered during previous crises, which has decreased significantly in the last months of the sample (Figure 12).

We find it necessary to justify the significant error in adjusting the estimates of expected growth during the COVID-19 pandemic. First, unlike previous recessions, this crisis originated from non-economic factors related to a health emergency, making it harder for policymakers to gauge the impact of shocks and increasing uncertainty around the adequacy of policy responses. Second, although the pandemic had widespread global repercussions, its effects were not uniform, varying across countries and sectors, which disrupted supply chains, particularly in manufacturing.

5. Conclusions

The analysis of the relationship between uncertainty and expected growth is of interest both theoretically and empirically, as well as for economic policy. Policymakers need to understand the channels through which uncertainty impacts the real economy since high uncertainty is associated with growth at risk. This topic has been of special interest since COVID-19 pandemic, a shock that has affected global economic activity, labor markets, and financial markets, generating a high degree of uncertainty about the evolution of the economy and the possible long-term consequences across countries.

In this paper, we examined the empirical relationship between uncertainty and growth expectations in Mexico. First, we analyzed the impact of uncertainty on growth using a VAR model for IGAE annual growth rates, controlling for the MUI and other factors such as the United States industrial production, exchange rate, inflation, consumption, investment, CETES rate, and financial conditions (measured by the FCI from Carrillo and García, 2021). Our analysis shows that uncertainty has a negative and statistically significant impact on growth, consumption, and investment in Mexico, and that increased uncertainty tightens financial conditions. In contrast, a shock to financial conditions has no significant effect on economic activity, consumption, or investment. Our results suggest that, once uncertainty is accounted for, financial conditions have limited power in explaining the dynamics of economic activity in Mexico.

Although the above results provide evidence that uncertainty is an important determinant of growth, Adrian *et al.* (2019) argue that point forecast estimates often ignore the bias of expectations around a central scenario and may therefore be overly optimistic. We thus analyze the impact of uncertainty on the distribution of the expected growth of economic activity in Mexico through the GaR methodology. Our results suggest that uncertainty has a negative and statistically significant impact on the left tail of the estimated distribution of expected economic growth in Mexico. Likewise, we find that an increase in uncertainty increases the dispersion and shifts the estimated distribution of expected growth to the left, which implies an increase in the probability of observing lower levels of growth. The preceding implies that uncertainty significantly contributes to explaining the negative bias of growth expectations.

Our estimates of the IGAE expected growth distribution, conditioned on the MUI and other economic factors, reveal that during periods of deep recessions, observed economic growth tends to align more closely with the lower quantiles of the distribution. For example, during the COVID-19 pandemic, the high levels of uncertainty created a pronounced negative bias in growth expectations. In this context, our GaR model using the MUI provides a more accurate conditional distribution of expected output growth, as it better reflects the realized growth rates. This is because heightened macroeconomic uncertainty appears to amplify the negative bias in growth projections, making central estimates overly optimistic in times of crisis.

Our results are robust to the inclusion of measures of financial conditions, uncertainty, and risk exposure in Mexico. In this regard, it stands out that, once we control for the MUI, financial conditions do not significantly affect the distribution of growth. We argue that measures of financial conditions for countries like Mexico could, in part, be a proxy for the level of uncertainty, given the greater exposure to risk and volatility of the economy, so that once there is an adequate measure of the uncertainty such as the MUI, financial conditions lose explanatory power.

Finally, we study alternative measures of economic activity on the demand, supply, and labor market sides. Specifically, indicators of consumption, investment, industrial production, the service sector, the level of employment, and the employment rate. Our baseline results hold for these alternative measures, although they do reveal slightly differentiated effects of uncertainty on their growth distributions. In particular, investment presented the greatest impact of uncertainty, while the employment rate registered the least impact. In

this way, a relevant extension of our analysis could consist of analyzing the impact of uncertainty on the distribution of expected growth for indicators listed above and others, possibly at the sectoral level, to identify the risks of growth expectations and their relationship with uncertainty.

References

- Adrian, T., N. Boyarchenko, and D. Giannone. 2019. Vulnerable growth, *American Economic Review*, 109(4): 1263-1289.
- Aquino, J., N. Ramírez Roldán, and L. Yépez. 2022. Does uncertainty matter for the effectiveness of monetary policy?, paper presented at the 2022 CEMLA-FRBNY-ECB Conference, Mexico City.
- Baker, S., N. Bloom, and S. Davis. 2016. Measuring economic policy uncertainty, *Quarterly Journal of Economics*, 131(4): 1593-1636.
- Baker, S., N. Bloom, S. Davis, and S. Terry. 2020. The impact of the recent spike in uncertainty on economic activity in the Euro area, NBER Working Paper No. 26983.
- Banxico. 2019. Crecimiento en riesgo y condiciones financieras en México, <https://www.banxico.org.mx/publicaciones-y-prensa/reportes-sobre-el-sistema-financiero/recuadros/%7BC724491B-E2B2-093B-AA2B-4635B91E9C73%7D.pdf>
- Banxico. 2020a. Condiciones financieras y crecimiento en riesgo ante el choque de COVID-19, <https://www.banxico.org.mx/publicaciones-y-prensa/reportes-sobre-el-sistema-financiero/recuadros/%7B62C55953-2958-DF13-F7E1-0227BA92338E%7D.pdf>
- Banxico. 2020b. El impacto de la incertidumbre sobre el consumo y la inversión en México, Informe Trimestral Octubre-Diciembre 2019, <https://www.banxico.org.mx/publicaciones-y-prensa/informes-trimestrales/recuadros/%7B63899F93-619D-0F97-7652-6833AF464F5C%7D.pdf>
- Bloom, N. 2009. The impact of uncertainty shocks, *Econometrica*, 77(3): 623-685.
- Bloom, N. 2014. Fluctuations in uncertainty, *Journal of Economic Perspectives*, 28(2): 153-176.
- Bonciani, D. and J. Oh. 2019. The long-run effects of uncertainty shocks, Bank of England Staff Working Paper No. 802.
- Bondell, H., B. Reich, and H. Wang. 2010. Noncrossing quantile regression by coupling quantiles, *Biometrics*, 66(4): 1090-1098.
- Carrillo, J. and A. García. 2021. The COVID-19 economic crisis in Mexico through the lens of a financial conditions index, Working Paper No. 2021-23, Banco de México.
- Catalán, H. 2019. Uncertainty and its impact on the Mexican economy, *Economía Coyuntural*, 4(4): 41-67.
- Cebros, A., A. Salcedo, D. Chiquiar, and A. Heffner-Rodríguez. 2020. Trade policy uncertainty and its effect on foreign direct investment: Evidence from Mexico, Working Paper No. 2020-14, Banco de México.

- ECB. 2016. The impact of uncertainty on activity in the euro area, <https://www.ecb.europa.eu/press/economic-bulletin/html/eb201608.en.html>
- Gieseck, A. and S. Rujin. 2020. The impact of the recent spike in uncertainty on economic activity in the euro area, https://www.ecb.europa.eu/press/economic-bulletin/focus/2020/html/ecb.ebbox202006_04~36366efeb.en.html
- Gu, X., X. Cheng, Z. Zhu, and X. Deng. 2021. Economic policy uncertainty and Chinas growth-at-risk, *Economic Analysis and Policy*, 70: 452-467.
- Hatzius, J. and S.J. Stehn. 2018. The case for a financial conditions index, Global Economics Papers, Goldman Sachs.
- Jovanovic, B. and S. Ma. 2020. Uncertainty and growth disasters, NBER Working Paper No. 28024.
- Jurado, K., S.C. Ludvigson, and S. Ng. 2015. Measuring uncertainty, *American Economic Review*, 105(3): 1177-1216.
- Koenker, R. and G. Bassett. 1978. Regression quantiles, *Econometrica*, 46(1): 33-50.
- Koop, G. and D. Korobilis. 2014. A new index of financial conditions, *European Economic Review*, 71: 101-116.
- Lloyd, S., G. Mantoan, and E. Manuel. 2022. When growth-at-risk hits the fan: Comparing quantile-regression predictive densities with committee fan charts, Unpublished.
- López, G. and G. Bush. 2019. Uncertainty and exchange rate volatility: The case of Mexico, Working Paper No. 2019-12, Banco de México.
- Plagborg-Møller, M., L. Reichlin, G. Ricco, and T. Hasenzagl. 2020. When is growth at risk?, *Brookings Papers on Economic Activity*, 2020: 1-81.
- Prasad, A., S. Elekdag, P. Jeasakul, R. Lafarguette, A. Alter, A.X. Feng, and C. Wang. 2019. Growth at risk: Concept and application in IMF country surveillance, IMF Working Paper No. 19-36.